

Visuals (Code + Plots)

Most Similar Words to “great”

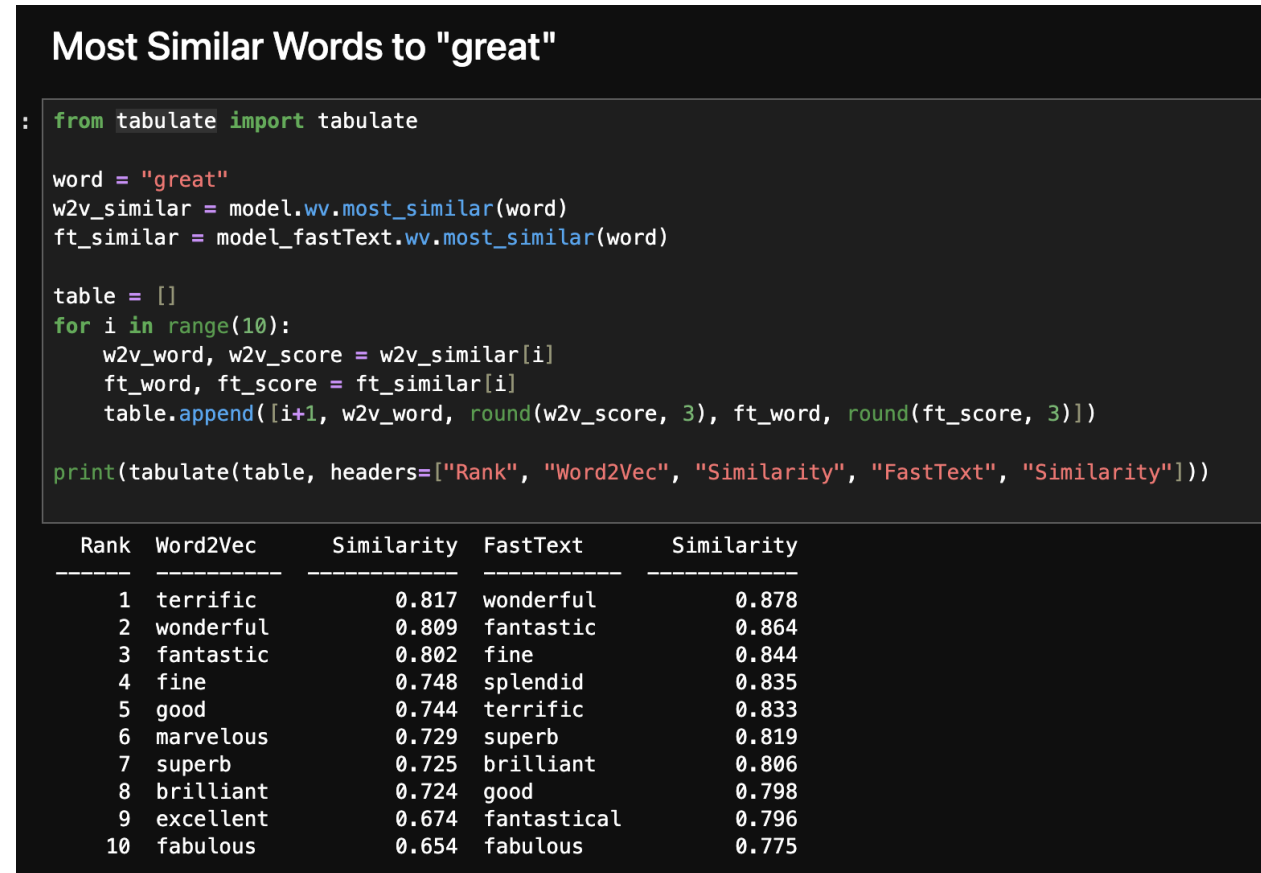


Figure 1. Comparison of the top 10 most similar words to “great” using Word2Vec and FastText embeddings.

While both models capture clear positive sentiment, FastText shows higher similarity scores and includes morphological variants like “fantastical”, reflecting its ability to model subword structure.

PCA Plots (Word2Vec vs. FastText)

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```
]: from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Select 100 shared words
words = list(model.wv.index_to_key)[:100]
w2v_vectors = [model.wv[word] for word in words]
ft_vectors = [model_fastText.wv[word] for word in words]

w2v_pca = PCA(n_components=2).fit_transform(w2v_vectors)
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.title("Word2Vec PCA")
for i, word in enumerate(words):
    plt.scatter(*w2v_pca[i])
    plt.annotate(word, w2v_pca[i], fontsize=8)

ft_pca = PCA(n_components=2).fit_transform(ft_vectors)
plt.subplot(1, 2, 2)
plt.title("FastText PCA")
for i, word in enumerate(words):
    plt.scatter(*ft_pca[i])
    plt.annotate(word, ft_pca[i], fontsize=8)

plt.tight_layout()
plt.show()
```

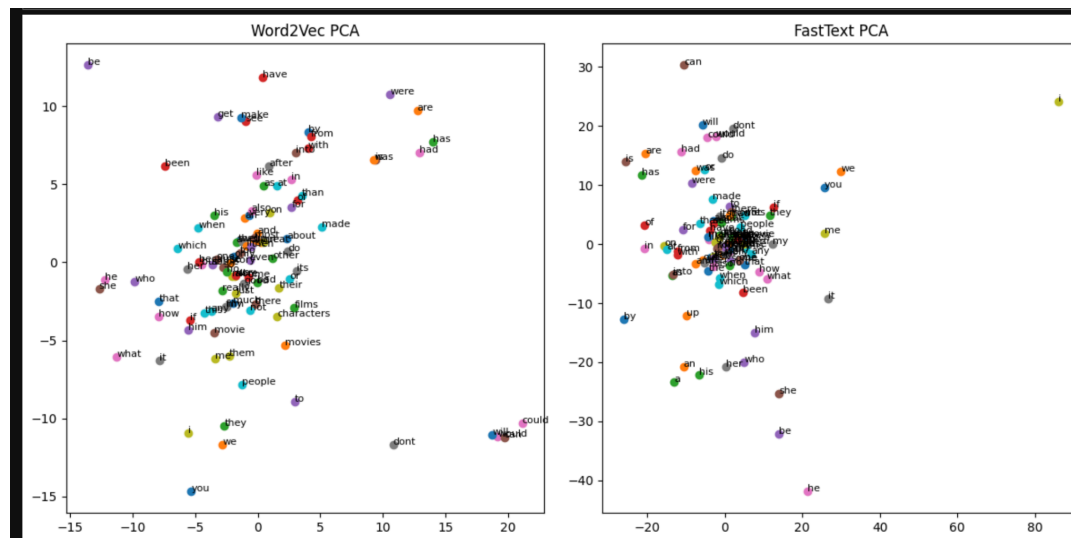


Figure 2. PCA projection of 100 shared words from Word2Vec and FastText embeddings.

Word2Vec exhibits a broader spatial spread, with loosely formed clusters, whereas FastText yields a more compact distribution. This reflects FastText's ability to generalize across subword patterns, often compressing variations of similar words into tighter vector neighbourhoods.

Before / After Preprocessing

```
] raw_review = "I absolutely loved this film!! <br /> One of the best ever."
cleaned_review = raw_review.lower().replace("<br />", " ")
cleaned_review = re.sub(r"[^a-zA-Z0-9\s]", "", cleaned_review)
tokens = cleaned_review.split()

table = [
    ["Raw", raw_review],
    ["Cleaned", cleaned_review],
    ["Tokenized", str(tokens)]
]

print(tabulate(table, headers=["Step", "Output"], tablefmt="grid"))
```

Step	Output
Raw	I absolutely loved this film!! One of the best ever.
Cleaned	i absolutely loved this film one of the best ever
Tokenized	['i', 'absolutely', 'loved', 'this', 'film', 'one', 'of', 'the', 'best', 'ever']

Figure 3. Text preprocessing steps applied to a raw IMDB review.

The HTML tag (
) and punctuation are removed, the text is lowercased, and the sentence is tokenized into individual words. This format prepares the data for Word2Vec and FastText training.